Name of the startup company : OxyLora

OXYLORA

Project Team

Oxylora are team from Baqa al-Hatab Secondary School, the team consists of two students: Sewar Al-Sheikh Ali and Roaa Al-Barghouthi, we are classmates from the same class, we cooperated in designing the project model, assembling and installing the parts and tools, and writing the work plan, Sewar wrote the Arduino code due to her good experience with programming codes, and Roaa designed the circuit diagram and logo due to her good experience in the field of design, we are working on developing an integrated system for a smart window that opens automatically when harmful gases accumulate such as cooking gas and CO2 to purify the air and prevent suffocation accidents and fires, and we plan to activate the project in homes in particular and in any place threatened by the danger of these toxic gases.

Project Introduction

Our product is an integrated electromechanical system that seeks to solve the problem of toxic gas accumulation in homes caused by heating and cooking gas, which is the cause of many cases of suffocation and fires and its serious harm to the elderly, children, and asthma patients. It is installed in the small upper parts of kitchens and rooms of the house and aims to improve home safety. The system consists of various gas sensors, such as the MQ-6 sensor, which is an electronic sensor used to detect flammable gases such as liquefied petroleum gas (LPG), and the MG811 sensor, which is used to detect CO2. It also includes a microcontroller (Arduino Uno) for signal processing and an electric motor (servo motor). The sensors measure the levels of toxic gases in the atmosphere and send the readings to the control unit. Once the gas levels in the room rise, the Arduino sends an electrical signal to the motor connected to the window, which opens until the air is replenished and then closes automatically. When the window is opened due to a leak and the gas level rises, the system sends an alert to the homeowner, either by phone or by text message, informing them of a problem at the source of the buildup that must be resolved. The system is designed to be compact, low-cost to install, and suitable for every home that relies on gas. This detects gas buildup early, preventing it from worsening, and also raising awareness of the dangers of gases.

Oxylora’s Project Summary:

In a world striving for development and finding solutions to all the problems facing humans, we found that there are a number of people who suffer from the risk of suffocation due to the accumulation of toxic gases such as CO2 due to the closure of ventilation openings such as windows, especially in winter, as well as the leakage of cooking gas without the residents noticing, which leads to suffocation and fires. Here comes the role of our system, which consists of a group of parts that work to perform an integrated function. The first is the QM-6 cooking gas sensor that measures the percentage of cooking gas and the CO2 sensor that measures the percentage of CO2. It reads the percentage of gases and sends them in the form of electrical signals to the Arduino Uno control unit, which reads the signals sent by the sensors. If the percentage of gas exceeds the required limit, it will send an electrical signal to the servo motor connected to the window to open automatically until the room is ventilated If the window is opened, the Arduino will contact the homeowner to inform him of the presence of danger and work to find a solution to the problem.

Value Proposition

After reviewing the solutions available from competing companies, we found that the available products either detect gas without automatically taking action, or they extract air without introducing clean air and oxygen into the space. These companies offer their products at a very high cost, which not everyone can afford. What distinguishes our product most is:

1. Our product relies on a set of electronic components available in all markets around the world, making it applicable in any country.
2. Our project operates automatically, as the window opens and closes automatically without the need for human intervention.
3. It provides peace of mind for the user, saving lives from the possibility of suffocation, fires, or fatalities, especially when children and the elderly are left in the home, car, or any closed room.
4. Our product alerts the homeowner via a message when the window is opened, enabling the homeowner to resolve the problem before it escalates or recurs.
5. Our system can be integrated with smart home systems such as Alexa.

6- We will program an application so that it is linked to a robot, and when the cooking gas level rises, the leakage source is closed upon the homeowner’s command..

Storytelling

With the development witnessed by the world, all the requirements of life have become easy for people, as they have saved a lot of time and effort on the one hand, but exposed them to danger on the other hand. Therefore, this danger must be reduced. After we conducted a number of studies, we found that there are large numbers of people exposed to the danger of gases. With the intensification of cold weather, several groups face difficulties in providing warmth, especially those living in mountainous areas where the temperature reaches below zero. In these areas, death accidents are frequent due to the misuse of various heating methods or their failure to comply with safety standards. Also, closing rooms and the lack of ventilation leads to the risk of suffocation and signs of fire, producing CO2 and many toxic gases that take a higher percentage than O2. From our personal experience in our community, we have heard of many cases of fainting and burning in closed places, which led to material and human losses. Our failure to find a product available in the local market with the same features as ours at an appropriate price and high quality was a real motivation to invent this idea and work on this project despite the challenges.

Environmental/Social impact

A new study relates 6000 annual deaths in Eastern Europe to heating systems	https://eeb.org/as-much-as-car-accidents-in-western-europe-new-study-links-6000-annual-deaths-in-eastern-europe-to-heating-and-cooking-systems/
Why cooking causes 4 million premature deaths every year?	https://www.economist.com/graphic-detail/2024/07/12/why-cooking-causes-4m-premature-deaths-a-year
The effect of cooking gas on air quality	https://www.eatingwell.com/is-cooking-on-a-gas-stove-affecting-the-air-quality-in-your-home-8731670?utm_source=
Indoor gas cooking and respiratory symptoms in the ECRHS cluster	https://pubmed.ncbi.nlm.nih.gov/38183794/
Cooking gas is the leading cause of home fires and injuries. It is also the second most common cause of death in home fires.	https://www.margatefl.com/836/Cooking-Fires

Target Market

1. Our project specifically targets the elderly, children, and those with respiratory illnesses, as they are most affected by gas buildup.
2. Residents of cold countries that rely on heating technologies and suffer from gas buildup.

We will develop a robust marketing plan aimed at raising awareness of the importance of implementing our project in the community.

Channels

Currently, we aspire to communicate and present our idea to companies according to specific terms and conditions that preserve our ownership rights for this innovation. We will cooperate with them to hire an experienced team (programmer, maintenance team, marketing expert) in addition to developing a solid marketing plan that will enable us to reach the largest possible number of our target group. We aspire to create an application for our company that focuses more on weekly newsletters that will introduce people to our company and our idea and increase their awareness of the importance of implementing this project. When this idea succeeds and the company's name becomes prominent, we will invent more innovative products that will enable us to establish an actual headquarters for us.

competitors:

Traditional gas leak alarms (such as Kidde, First Alert, X-Sens) only issue an alert without taking automatic action. Furthermore, some of their products are expensive for some. In contrast, our project generates an alert that prompts a decision to open the window, along with a moderate cost. It also sends a message to the owner's phone.

We target :

*Residential homes, especially those with children and the elderly.

*In kitchens, bathrooms, and farms.

*Families that use gas heating or cooking.

Traditional alarms currently hold the largest share, but they are not interactive, and their standards are limited to a limited segment. Our project offers an ideal, smart, and effective solution at an affordable price, enabling us to capture a larger market share in the future.

Revenue Streams

We will generate income from our project primarily by selling the product to owners of areas at risk of gas accumulation, in addition to providing an annual subscription service for maintenance and periodic follow-up throughout the year, in addition to continuous and immediate notifications.

Cost Structure:

Fixed costs are the prices of electronic and electrical .components
Variable costs vary depending on the market (USA, Europe, .Palestine) and delivery costs

- In addition to our desire for this product to be of high standards, we will supervise:

- Safety and electrical certifications (e.g., CE, FCC).
- Heat and moisture resistant case .
- Home installation and user manual .

- When starting the project, we will purchase these tools directly from companies at wholesale prices, which reduces costs and increases profits.

20$	Arduino uno controller
35$	MHZ19B (CO2)sensor
10$	MQ6 (LPG) sensor
45$	Stepper Motor + Gearbox
10$	LCD with I2C
10$	SIM800L
10$	Wires and connections
10$	12v power converter with voltage regulator
7$	electronic protection box
≈ 160$	**Total costs**

About the Pilot Project

Over the next three months, we will begin the pilot phase of our system. We aim to test and verify the system's efficiency in identifying and detecting toxic gases, automatically opening the window, and sending a message to the homeowner. This phase includes the following steps:

1. Assembling the electronic components, such as the control unit and sensors.
2. Installing the electronic circuit based on its initial design.
3. We will write a program code for the system that specifies the conditions for opening and closing the window based on the gas level measured by the sensors.
4. Conducting a test on a small group of homes to gather feedback and refine the concept as needed.
5. Based on the results, we will modify and develop the programming and improve the connection of the components.

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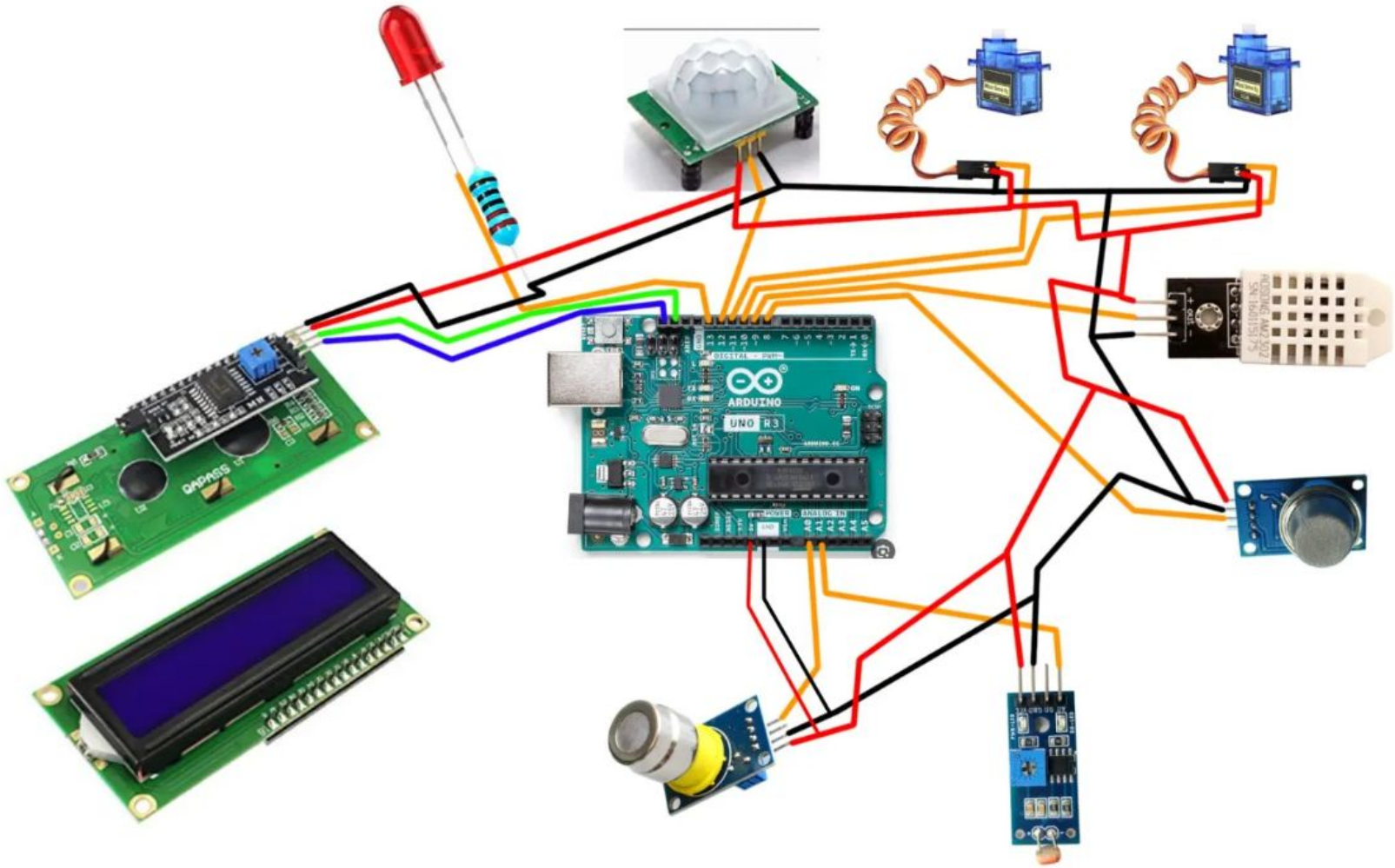
Presentation of the product or application for the project:

https://drive.google.com/drive/folders/1pzuIrKlq1rRcsCHaUiPzvM3Bk4EaHDaS?usp=sharing	Project photos
https://drive.google.com/drive/folders/1ffUwgjJ_5IEN8e1U2271PKhoneKwI5KO	Project video
https://www.instagram.com/oxy_lora/	Instagram page

Closing Statements

The Oxylora project is not just a technical idea, but rather a system through which we seek to protect lives and property from the danger of gas accumulation using innovative methods. We still see the need to implement it in all homes, thereby ensuring safety and peace of mind. That's why we aspire to gain the support of interested parties and investors who believe in the power of innovation to serve life, so that together we can transform a small idea into a real and tangible impact.

Circuit diagram:



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Smart Window Project Survey

We are developing a smart window that interacts with its surrounding environment. It will open automatically when it detects a cooking gas leak or an increase in carbon dioxide in the atmosphere, and will close after environmental conditions have stabilized. We would like your opinion on this idea and its effectiveness.

How do you rate the importance of having a device to control indoor air quality? *

- Very Important
- Somewhat Important
- Not Important

Do you think that the idea of a smart window that automatically opens and closes in the event of a cooking gas leak or an increase in CO2 could contribute to increased safety in the home? *

- Yes
- No
- Not Interested

Would you prefer the window to be controlled via a smartphone app or via automatic sensors? *

- Automatic control via sensors
- Control via a phone app
- I don't need any of these options

Or another HEPA filter. Would you prefer these windows to have additional features such as the ability to purify the air? *

- Yes
- No
- Rima

Gender *

Male 0

Female

What are the most important gases that you would like to know the percentage of in the air around you? (You can choose more than one answer.) *

- [ ] Carbon monoxide CO
- [ ] Carbon dioxide CO2
- [ ] Cooking gas

Do you think that cooking gas leaks or increased levels of carbon dioxide in homes pose a health risk? *

Yes

- No
- Not sure

Have you ever used air purifiers or any solutions to improve the air quality in your home? *

Yes

- Y
- Not interested

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* What feature do you value most about our smart window? You can choose more than one answer.

- [ ] Improving air quality
- [ ] Providing safety against harmful gases
- [ ] Saving energy
- [ ] Smart ventilation
- [ ] Sending a phone notification when pollution levels rise

* If you own our smart window, where would you prefer to put it? You can choose more than one answer.

- [ ] Bedroom
- [ ] Living room
- [ ] Office
- [ ] Kitchen
- [ ] Bathroom

Do you have any comments or suggestions regarding the design or functionality of the smart window?

Your answer

Submit

Clear form

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Gender

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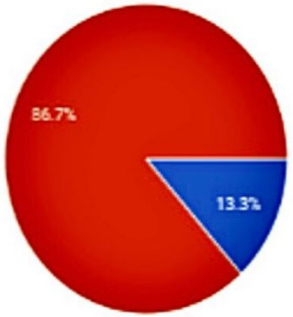

Male

female

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Do you think the idea of a smart window that opens and closes automatically in the event of a cooking gas leak or excess CO2 could contribute to increased home safety?

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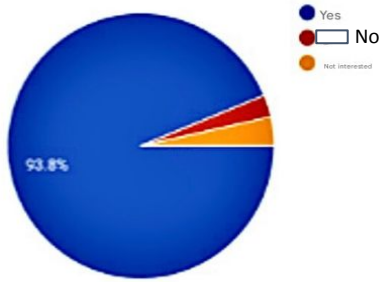

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Have you ever used air purifiers or any solutions to improve the air quality in your home?

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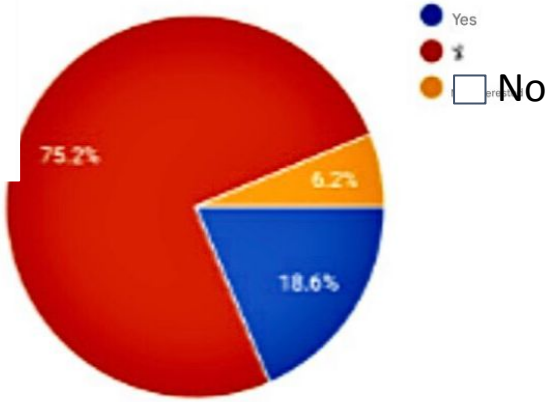

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How important do you rate the presence of an air quality control device inside homes?

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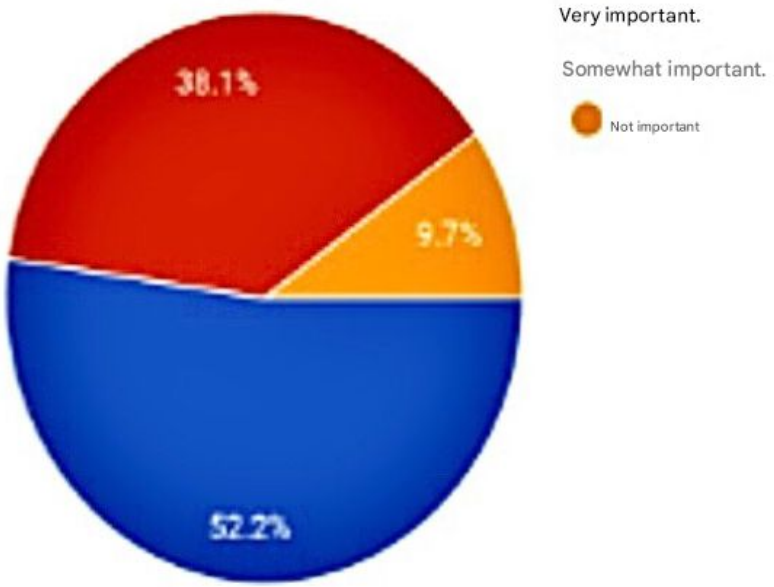

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Do you think that cooking gas leaks or increased carbon dioxide levels in homes pose a health risk?

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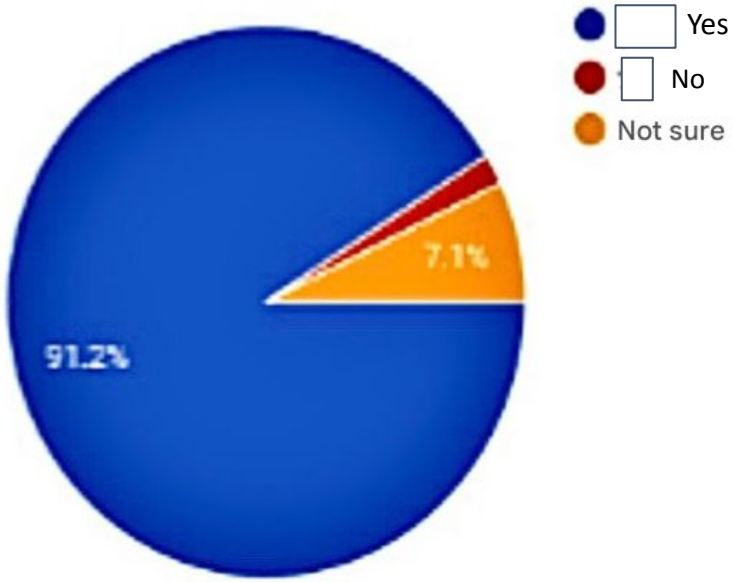

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Would you prefer the window to be controlled by a smartphone app or by automatic sensors?

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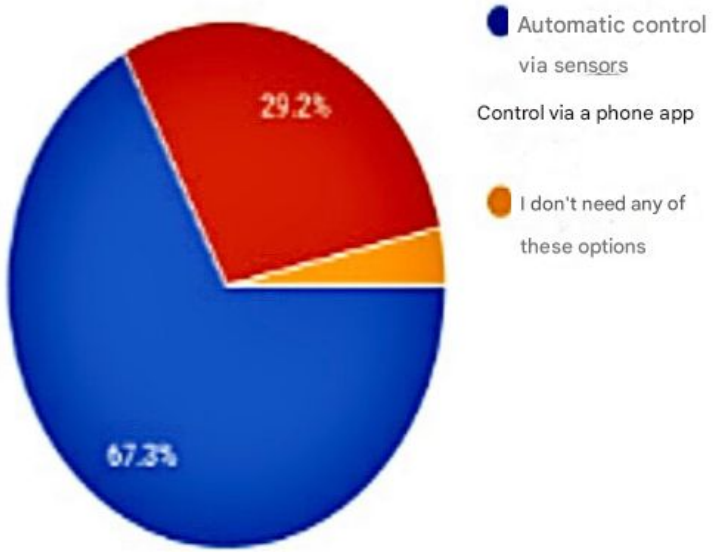

Would you prefer that versions of the chart these windows have additional features such as air purification capability (HEPA or other filter)?

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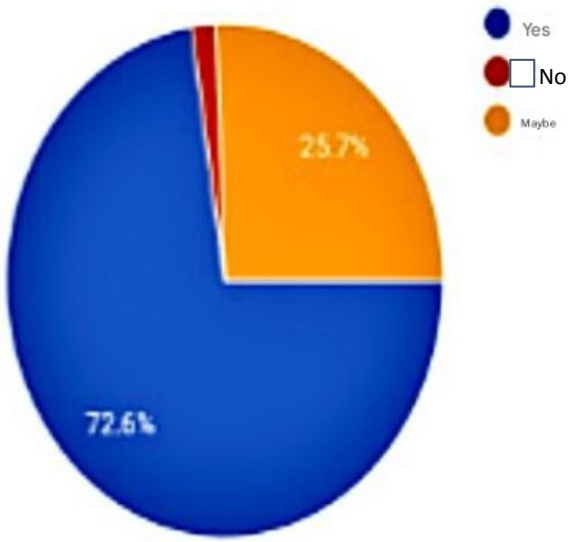

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What feature do you care about most in our smart window? (You can choose more than one answer)

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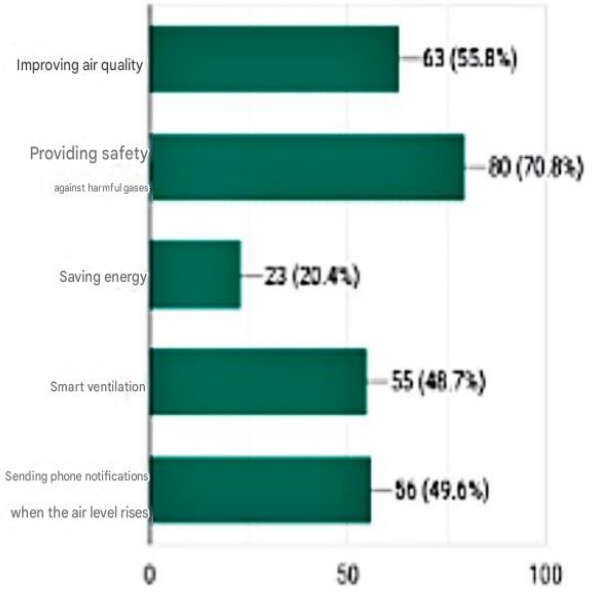

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What are the most important gases that you would like to know the percentage of in the air around you? (You may choose more than one answer)

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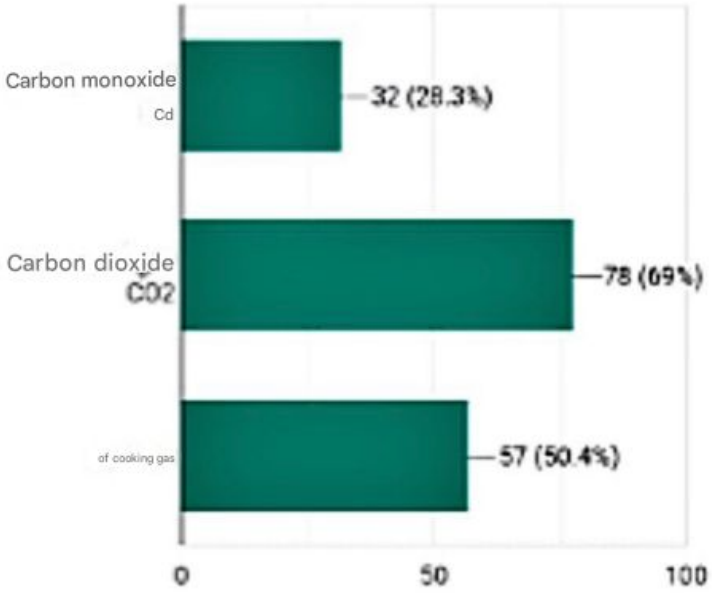

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If you had our smart window Where would you prefer to put it? (You can choose more than one answer)

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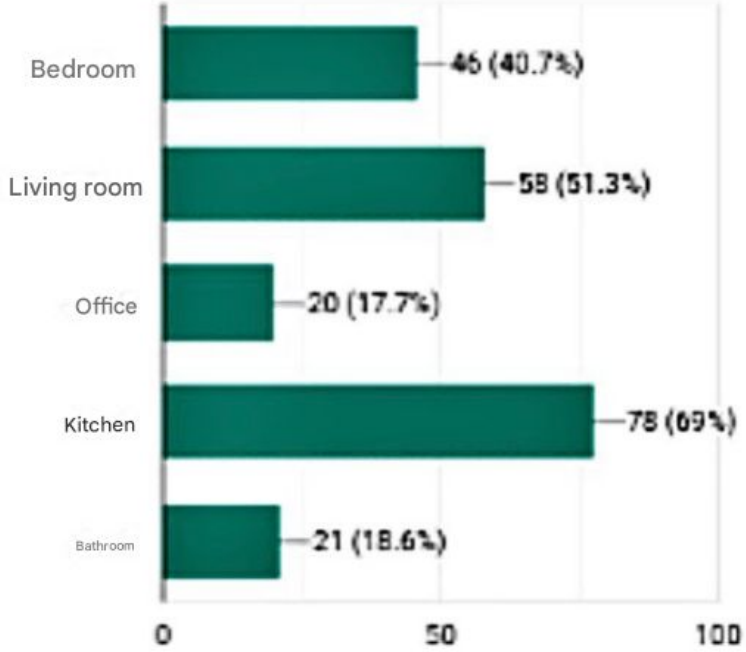

Financial Study

Total expected cost ($)	Expected number of units	Expected unit cost ($)	Unity	Item	#
				work team	1
18000	12	1500	monthly	Project Manager/Project Owner	1.1
6000	12	600	monthly	technical	1.2
24000		2100		**Subtotal (1)**	
				One-time purchases	2
20	1	20	yearly	Arduino Uno controller	2.1
45	1	45	yearly	gas sensors	2.2
10	1	10	yearly	LCD screen	2.3
10	1	10	yearly	SIM card	2.4
85		85		**Subtotal (2)**	
				Fixed monthly expenses	3
96	12	8	monthly	electricity	3.1
24	12	2	monthly	SIM service subscription	3.2
600	12	50	monthly	Cross-platform marketing campaigns for the first year	3.3
720		80		**Subtotal (3)**	
				Variable monthly expenses	4
70	1	160	yearly	Price of electronic parts	4.1
65	1	65	once	Additional shipping or installation fees apply to customers.	4.2
135		135		**Subtotal (4)**	
9580		1000		**Total**	

Business Model Canvas

Key Partners

1- Companies that manufacture sensors, motors and control units.

2- Suppliers and manufacturers of window and glass frames.

3-Programming companies.

4- Financial partnerships (investors).

Key Activities

1- Content marketing: We will write awareness articles such as "How to protect your family from gas leaks." At the end of the article, we will present our product as a solution to this problem.

2-Digital marketing.

3--Research and development of accurate gas sensors.

Key Resources

1- Physical resources (sensors, control unit)

2-Human resources (programmer, maintenance team).

3- Financial resources (initial capital).

4-Intellectual resources (project idea, intellectual property rights).

Value Proposition

1-Providing psychological comfort to the user by ensuring the safety of children, the elderly, and respiratory patients from any danger of gas accumulation.

2- Providing a life-saving product with diverse features at an affordable price..

3- Providing instant alerts via phone when the window is opened, and monthly bulletins on air quality and tips for improving it.

4-The system operates automatically without the need for human intervention.

Customer Relationship

1- Communicate with customers via email through weekly newsletters.

2- (5% cashback) for the customer who promotes our product and brings new customers.

3- Obtaining certificates that document the credibility and efficiency of our company's products.

Channels

1-The application is created by displaying the product with all its details (price, specifications, etc.) and recording all customer information. When the deal is completed, the customer is contacted via WhatsApp to learn the details.

2- Promoting our product in exhibitions and local gatherings.

3- Promoting our product through social media and sponsored ads.

Customer Segments

1-Homes at risk of gas buildup

2- Patients with respiratory diseases

3- Children and the elderly

Cost Structure

1-Development costs (system testing and design improvement)

2- Fixed costs (employee salaries).

3-Variable costs (price of electronic parts)

4- Marketing and distribution costs (campaigns on Instagram or Facebook and delivering the product to customers)

Revenue Streams

1- Selling the product primarily

2- Technical support and maintenance

3- Annual subscription.

Social impact

Protection from suffocation and fire accidents.

Environmental impact

Improve air quality, reduce household environmental hazards.